WORKED SOLUTION

STEP II 1991, Question 5

Pure • Integration Reduction Formulae

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s2-q5

The key idea

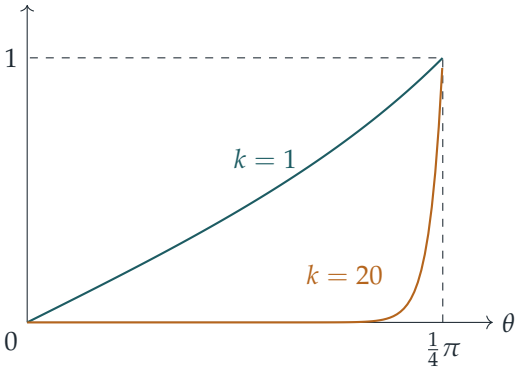
Write $I_k = \int_0^{\pi/4} \tan^k \theta \, \mathrm{d}\theta$. Odd and even powers of tan don't integrate directly, but two of them added together do: $\tan^k \theta + \tan^{k+2} \theta = \tan^k \theta \sec^2 \theta$, which is the derivative of $\tan^{k+1} \theta / (k+1)$. That reduction formula, unwound by induction, gives the formula; the sketch for large k tells you the integral shrinks to zero, and that is what turns the formula into the sum of an infinite series.

Solution

The sketches

On $[0, \frac{1}{4}\pi]$ we have $0 \leq \tan\theta \leq 1$, with $\tan 0 = 0$ and $\tan \frac{1}{4}\pi = 1$. For $k = 1$ the graph rises from $(0,0)$ to $(\frac{1}{4}\pi, 1)$, gently curving upwards (it starts with gradient 1 and ends with gradient $\sec^2 \frac{1}{4}\pi = 2$).

For $k \gg 1$, every value of $\tan\theta$ below 1 is crushed towards 0 by raising it to a high power, while the end values 0 and 1 stay put. So $\tan^k \theta$ hugs the axis for most of the interval and then shoots up to 1 very close to $\theta = \frac{1}{4}\pi$. In fact, writing $\theta = \frac{1}{4}\pi - u$ with u small, $\tan\theta \approx 1 - 2u$, so $\tan^k \theta \approx (1-2u)^k \approx \mathrm{e}^{-2ku}$: the spike has width about $1/(2k)$, and the area under it is about $1/(2k)$.



The reduction formula

For $k \geq 0$,

$$I_k + I_{k+2} = \int_0^{\pi/4} \tan^k\theta\,(1+\tan^2\theta)\,\mathrm{d}\theta = \int_0^{\pi/4} \tan^k\theta\sec^2\theta\,\mathrm{d}\theta = \Big[\frac{\tan^{k+1}\theta}{k+1}\Big]_0^{\pi/4} = \frac{1}{k+1}.$$

We also need the two starting values:

$$I_0 = \tfrac{1}{4}\pi, \qquad I_1 = \int_0^{\pi/4} \frac{\sin\theta}{\cos\theta}\,\mathrm{d}\theta = \Big[-\ln\cos\theta\Big]_0^{\pi/4} = -\ln\tfrac{1}{\sqrt{2}} = \tfrac{1}{2}\ln 2.$$

The odd powers

Let $J_n = I_{2n+1}$. Taking $k = 2n-1$ in the reduction formula gives

$$J_n = \frac{1}{2n} - J_{n-1} \qquad (n \geq 1).$$

We prove the required formula by induction. For $n = 1$: $J_1 = \frac{1}{2} - J_0 = \frac{1}{2} - \frac{1}{2}\ln 2 = (-1)^1(\frac{1}{2}\ln 2 - \frac{1}{2})$, which is the formula with $n = 1$.

Suppose it holds for $n - 1 \geq 1$. Then

$$\begin{aligned} J_n &= \frac{1}{2n} - (-1)^{n-1}\Big(\tfrac{1}{2}\ln 2 + \sum_{m=1}^{n-1}\frac{(-1)^m}{2m}\Big) \\ &= (-1)^n\Big(\tfrac{1}{2}\ln 2 + \sum_{m=1}^{n-1}\frac{(-1)^m}{2m}\Big) + (-1)^n\,\frac{(-1)^n}{2n} \\ &= (-1)^n\Big(\tfrac{1}{2}\ln 2 + \sum_{m=1}^{n}\frac{(-1)^m}{2m}\Big), \end{aligned}$$

using $(-1)^n(-1)^n = 1$ to write $\frac{1}{2n}$ as the $m = n$ term inside the bracket. By induction the formula holds for every positive integer n, as required.

The first series

Ask yourself How do we know $J_n \to 0$, rather than just "it looks small on the sketch"?

We need an honest bound. On $[0, \frac{1}{4}\pi]$ we have $\tan\theta \geq 0$ and $\sec^2\theta \geq 1$, so

$$0 \leq J_n = \int_0^{\pi/4} \tan^{2n+1}\theta \,\mathrm{d}\theta \leq \int_0^{\pi/4} \tan^{2n+1}\theta \sec^2\theta \,\mathrm{d}\theta = \frac{1}{2n+2}.$$

Hence $J_n \to 0$ as $n \to \infty$, which is exactly what the large-k sketch suggested (and the bound $1/(2n+2)$ matches the spike's area of about $1/(2k)$). Since $|J_n|$ is the absolute value of the bracket in the formula,

$$\left|\tfrac{1}{2}\ln 2 + \sum_{m=1}^{n} \frac{(-1)^m}{2m}\right| = J_n \leq \frac{1}{2n+2} \to 0,$$

so the partial sums $\sum_{m=1}^{n} \frac{(-1)^m}{2m}$ converge to $-\frac{1}{2}\ln 2$. Multiplying by -1,

$$\sum_{m=1}^{\infty} \frac{(-1)^{m-1}}{2m} = \tfrac{1}{2}\ln 2.$$

The second series

Now do the same with even powers. Let $K_n = I_{2n}$. Taking $k = 2n-2$ in the reduction formula gives

$$K_n = \frac{1}{2n-1} - K_{n-1} \qquad (n \geq 1), \qquad K_0 = \tfrac{1}{4}\pi.$$

Exactly the same induction (now starting from $K_1 = 1 - \frac{1}{4}\pi$) gives

$$K_n = \int_0^{\pi/4} \tan^{2n}\theta \,\mathrm{d}\theta = (-1)^n\left(\tfrac{1}{4}\pi + \sum_{m=1}^{n} \frac{(-1)^m}{2m-1}\right):$$

the inductive step is $K_n = \frac{1}{2n-1} - (-1)^{n-1}(\cdots) = (-1)^n(\cdots) + (-1)^n\frac{(-1)^n}{2n-1}$, just as before. The same bound gives

$$0 \leq K_n \leq \int_0^{\pi/4} \tan^{2n}\theta \sec^2\theta \,\mathrm{d}\theta = \frac{1}{2n+1} \to 0,$$

so $\sum_{m=1}^{n} \frac{(-1)^m}{2m-1} \to -\frac{1}{4}\pi$, that is,

$$\sum_{m=1}^{\infty} \frac{(-1)^{m-1}}{2m-1} = 1 - \frac{1}{3} + \frac{1}{5} - \cdots = \frac{\pi}{4}.$$

Checks

- With $n = 1$: $\int_0^{\pi/4} \tan^3\theta \,\mathrm{d}\theta = \frac{1}{2} - \frac{1}{2}\ln 2 \approx 0.15343$, and Simpson's rule on the integral gives the same to ten decimal places. We checked the formulas for I_{2n+1} and I_{2n} this way for $n = 1, \ldots, 11$.

- For large k, numerically $I_{100} \approx 0.0049995$ and $I_{1000} \approx 0.00049999996$, very close to $1/(2k)$, as the sketch predicts.
- A million terms of the two series give $0.346573\ldots$ and $0.785398\ldots$, against $\frac{1}{2}\ln 2 = 0.346574$ and $\frac{1}{4}\pi = 0.785398$. The error is about the size of the next term, as the bounds above suggest.

Common slips

- Saying "$\tan^{2n+1}\theta \to 0$, so the integral tends to 0" without a bound. The integrand does *not* tend to 0 at $\theta = \frac{1}{4}\pi$, so a reason such as the $\sec^2\theta \geq 1$ bound is needed.
- Sign errors in the induction: keep $(-1)^n$ outside the bracket and check the case $n = 1$ against a direct calculation.

Beyond STEP

The first series is half of the alternating harmonic series $1 - \frac{1}{2} + \frac{1}{3} - \cdots = \ln 2$, and the second is the Leibniz series for π. Both converge only *conditionally*: rearranging their terms can change the sum, a striking fact (Riemann's rearrangement theorem) met early in a first analysis course. The argument here, an exact error term that is squeezed to zero, is the same idea as Taylor's theorem with remainder. Abbott's *Understanding Analysis* treats both carefully.

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